

● EASY

● ADVANCED MATH

Advanced Math

03

10 questions · ~15 min

Q1

ADVANCED MATH

$$k^2 - 53 = 91$$

What is the positive solution to the given equation?

- A. 144
- B. 72
- C. 38
- D. 12

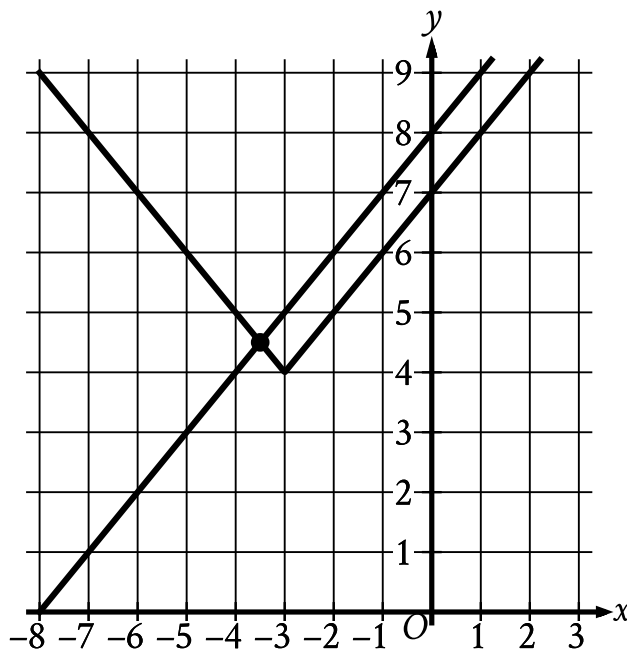
Which expression is equivalent to $(2x^2 - 4) - (-3x^2 + 2x - 7)$?

- A. $5x^2 - 2x + 3$
- B. $5x^2 + 2x - 3$
- C. $-x^2 - 2x - 11$
- D. $-x^2 + 2x - 11$

$$y = -\frac{1}{4}x^2 + 2x + 29$$

The given equation models a company's scheduled deliveries over 8 months, where y is the estimated number of scheduled deliveries x months after the end of May 2012, where $0 \leq x \leq 8$. Which statement is the best interpretation of the y -intercept of the graph of this equation in the xy -plane?

- A. At the end of May 2012, the estimated number of scheduled deliveries was 0.
- B. At the end of May 2012, the estimated number of scheduled deliveries was 29.
- C. At the end of June 2012, the estimated number of scheduled deliveries was 0.
- D. At the end of June 2012, the estimated number of scheduled deliveries was 29.



- For the absolute value function in the system:
 - Moving from left to right:
 - The function slants sharply down to the point (negative 3 comma 4).
 - The function then slants sharply up.
 - The function passes through the following points:
 - (negative seven halves comma nine halves)
 - (negative 3 comma 4)
 - (negative five halves comma nine halves)
- For the linear function in the system:
 - The function slants sharply up from left to right.
 - The function passes through the following points:
 - (negative seven halves comma nine halves)
 - (0 comma 8)

The graph of a system of an absolute value function and a linear function is shown. What is the solution x, y to this system of two equations?

A. 0,8

B. $\frac{7}{2}, \frac{9}{2}$

C. $-\frac{7}{2}, \frac{9}{2}$

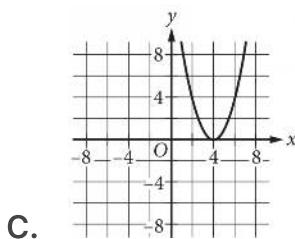
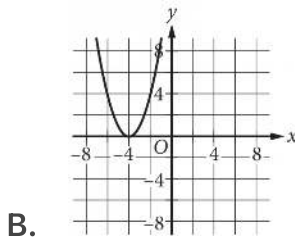
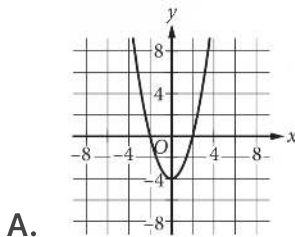
D. -3,4

Which of the following expressions is equivalent to the sum of $(r^3 + 5r^2 + 7)$ and $(r^2 + 8r + 12)$?

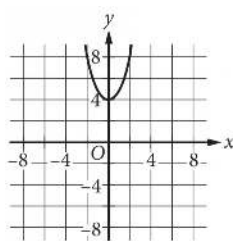
- A. $r^5 + 13r^3 + 19$
- B. $2r^3 + 13r^2 + 19$
- C. $r^3 + 5r^2 + 7r + 12$
- D. $r^3 + 6r^2 + 8r + 19$

$$f(x) = x^2 + 4$$

The function f is defined as shown. Which of the following graphs in the xy -plane could be the graph of $y = f(x)$?



D.



The total revenue from sales of a product can be calculated using the formula $T = PQ$, where T is the total revenue, P is the price of the product, and Q is the quantity of the product sold. Which of the following equations gives the quantity of product sold in terms of P and T ?

A. $Q = \frac{P}{T}$

B. $Q = \frac{T}{P}$

C. $Q = PT$

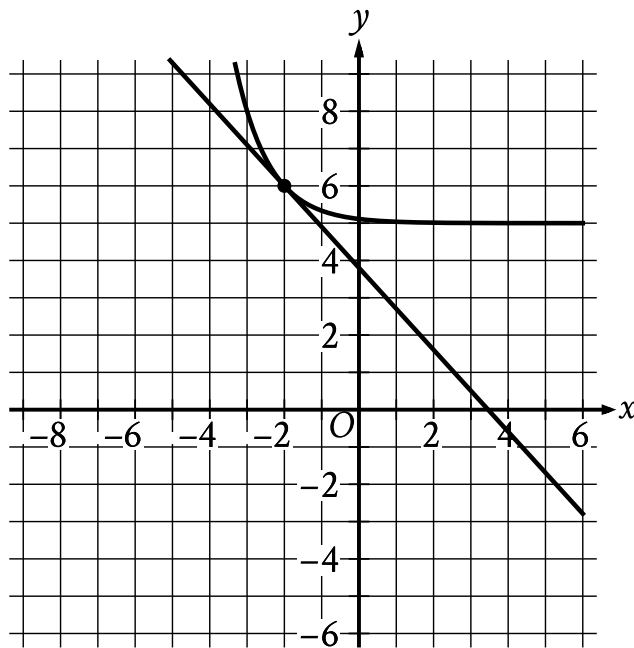
D. $Q = T - P$

Which expression is equivalent to $16x^3y^2 + 14xy$?

- A. $2xy8xy + 7$
- B. $2xy8x^2y + 7$
- C. $14xy2x^2y + 1$
- D. $14xy8x^2y + 1$

The function f is defined by $fx = 5x^2$. What is the value of $f8$?

- A. 40
- B. 50
- C. 80
- D. 320



- For the line in the system:
 - The line slants sharply down from left to right.
 - The line passes through the following points:
 - $(-2, 6)$
 - approximately $(0, 3.8)$
- For the curve in the system:
 - Moving from left to right:
 - The curve passes from quadrant 2 to quadrant 1.
 - In quadrant 2:
 - The curve trends down sharply to the point $(-2, 6)$.
 - The curve then trends down gradually to the y axis.
 - In quadrant 1, the curve trends down gradually.
 - As x increases, the curve approaches the line y equals 5.
 - The curve passes through the following points:
 - $(-3, 8)$
 - $(-2, 6)$

The graph of a system of a linear equation and a nonlinear equation is shown. What is the solution x, y to this system?

- A. 6, 0
- B. -2, 6
- C. 0, -2
- D. 0, 0